GS EXTERIOR EXPERTS · DECK DIVISION · MATERIAL TAKE-OFF + BUILD SET · FOR DECKS & DOCKS

# 281 S Jefferson Ave · Decks + Roof Covers

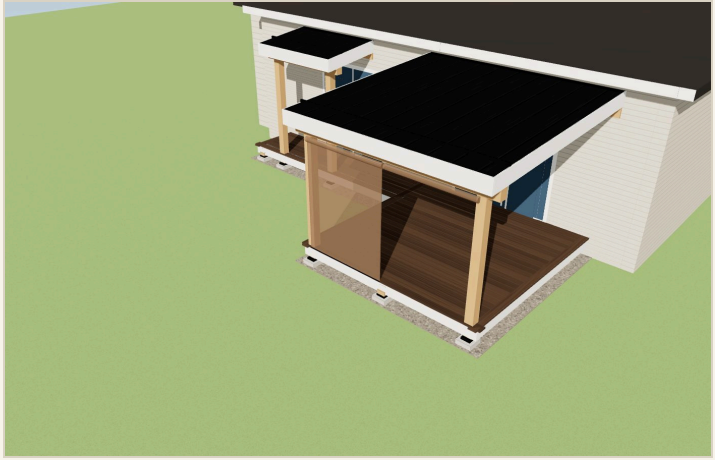
281 S Jefferson Ave, Louisville · two decks (12' x 12' + 14'8" x 3'8" walkway) on CEMCO steel and Diamond Piers · TimberTech Landmark American Walnut, picture frame + dividers, no butt joints · two lean-to covers, black standing seam, Hardie soffit / fascia / rakes · blinds kept · October 16, 2026

Account GSEXT1 · ship to 281 S Jefferson Ave, Louisville, CO 80027 · please quote every line; call Jade Pender 303-550-9558 with substitutions.

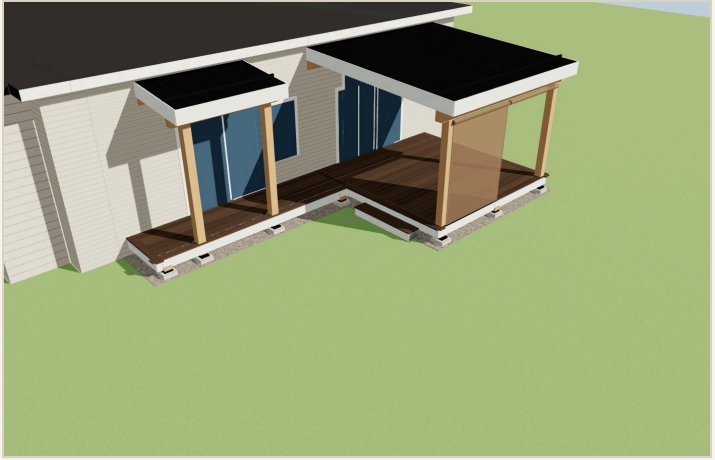
| SQUARE FEET | FOOTINGS | ORDER LINES | HEIGHT |
|---|---|---|---|
| **197.8** | **7** | **78** | **1'-3"** |

## THE JOB

| | |
|---|---|
| Deck | 197.8 SF · 1'-3" above grade · 2 zones, 26'-8" along the house, 12'-0" at the deepest |
| Joists | CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) @ 12" OC |
| Rims | single-ply 600T track rims (G90) and 600T ledger track on a steel angle |
| Beams | FLUSH boxed 600S162-54 W: 14'-1 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 471 lb); FLUSH boxed 600S162-54 M: 11'-5 1/2", 3 posts at 15'-2" / 20'-8" / 26'-2" (worst 1,613 lb) |
| Footings / posts | 3 x 6x6 Douglas fir rough deck posts on steel beam saddles + 4 6x6 Douglas fir rough cover posts carrying the deck beam on DP-50/50 |
| Decking | TimberTech Landmark American Walnut — 25 rows parallel to the house, picture frame & dividers in Landmark American Walnut, 3/16" gaps, CONCEALoc |
| Rail | none |
| Stairs | main deck side, toward the walkway: 2 risers @ 7.50", 1 treads, 6 stringers, 5'-0" wide |



Isometric from the stair corner



Isometric from the open corner

**MATERIAL TAKE-OFF** NET = counted off the drawings · ORDER = what to pull and ship

## Lumber

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 347 | **364** | LF | CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) — joists, PF joists, blocking | +5% (cut to length by the supplier where possible) — 12" OC |
| 77 | **82** | LF | CEMCO 600T track (54 mil / 16 ga, G90) — rims (single ply, caps the joist ends) + ledger track | +5% (51 LF rim + 27 LF ledger) |
| 27 | **30** | LF | Steel ledger angle 3" x 3" 14 ga G90 — joist seats on the ledger track | +10% |
| 51 | **54** | LF | CEMCO 600S162-54 C, G90 — boxed beams (2 C toe-to-toe) | +5% (FLUSH boxed 600S162-54 W 14.1' · FLUSH boxed 600S162-54 M 11.5') |
| 4 | **4** | ea | Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier) | exact (one per shared post, (2) 1/2" thru-bolts + #12 screws into the beam) |
| 1 | **1** | ea | 6x6x8 Douglas fir rough — deck posts on the deck-only piers | exact (3 posts, about 6" each, 8 per 8') |
| 3 | **3** | ea | Simpson ABA66RZ post base on the Diamond Pier bolt — deck post | exact |
| 3 | **3** | ea | Galvanized steel beam saddle, 11 ga — boxed steel beam to the 6x6 post top | exact (one per deck post, thru-bolted to the post, #12 screws into the beam) |

## Footings

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 7 | **7** | ea | Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps) | exact — confirm stock |

## Hardware

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 56 | **62** | ea | Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam | +6 (2 per joist end where it frames into a beam or ledger) |
| 3 | **3** | box | #12-14 x 1" hex-washer self-drilling screw, Climaseal, 500 ct | 1470 needed (clips, track, boxed beams, post brackets) |

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **1** | box | Structural screw 1/4" x 3-1/2" through the ledger track into the house rim board (or per the engineer) | 40 needed — 2 rows staggered 16" OC |
| 8 | **8** | ea | Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2 equivalent | exact (2 near each end of every ledger) |
| 2 | **2** | ea | Cold-galvanizing spray (cut ends, field drilling) | lot |

### Flashing & waterproofing

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **1** | roll | Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims) | 27 LF of house wall with laps — nothing on a privacy wall |
| 3 | **3** | ea | Aluminum L-flashing 10' (over ledger + side rims) | house walls only |
| 1 | **1** | roll | 6-mil poly sheeting 10' x 100' — under the decks | 218 SF with laps |
| 1.3 | **1.5** | CY | 3/4" washed gravel, 2" over the poly — under the decks (delivered) | 218 SF x 2" |

### Decking

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 25 | **27** | ea | TimberTech Landmark American Walnut 1x6x12 Grooved | +2 (25 x 12' · 7 x 16' · panel 1 7 rows · panel 2 25 rows · every row one piece) |
| 7 | **9** | ea | TimberTech Landmark American Walnut 1x6x16 Grooved | +2 (25 x 12' · 7 x 16' · panel 1 7 rows · panel 2 25 rows · every row one piece) |
| 4 | **4** | ea | TimberTech Landmark American Walnut 1x6x16 Square Edge — borders & dividers | exact (border 14'-8",12'-0" 12'-1" · border 26'-8",0" 12'-1" · breaker DW/M 12'-1" · border 14'-8",3'-8" 8'-4 15/16") |
| 1 | **2** | ea | TimberTech Landmark American Walnut 1x6x20 Square Edge — borders & dividers | +1 (border 0",3'-8" 14'-9" + border 0",0" 3'-9") |

### Fasteners

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **1** | box | TimberTech CONCEALoc hidden fastener 500 ct (screws incl.) | 437 needed · 500 supplied |
| 1 | **1** | box | Starborn Cap-Tor xd 2-3/4" Epoxy American Walnut 350 ct | 284 screws needed (first row 28 · border 116 · fascia 92 · stairs 48) |

### Fascia

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 5 | **6** | ea | Deck skirt — HardieTrim 4/4 x 12 x 12' smooth ColorPlus Arctic White (wraps the double rim, in place of the composite fascia) | +1 (left fascia (W) 3'-8 1/2" · right fascia (W) 3'-8 1/2" · front fascia (W) 12'-0" · front fascia (W) 2'-9" · left fascia (M) 12'-0" · left fascia (M) 1/2" · right fascia (M) 12'-0" · right fascia (M) 1/2" · front fascia (M) 12'-0" · front fascia (M) 1") |
| 1 | **1** | box | Color-matched trim screws for HardieTrim, 100 ct (Arctic White) | 40 color-matched screws, pre-drilled, 2 per joist line |

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **1** | kit | ColorPlus touch-up kit, Arctic White | cut ends |

## Stairs

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **2** | ea | TimberTech Landmark American Walnut 1x6x12 Square Edge — treads (main deck side, toward the walkway stair) | +1 (1 treads x 2 boards @ 5'-0", 2 per board) |
| 1 | **1** | ea | TimberTech Landmark American Walnut riser board 7-1/4" x 12' (main deck side, toward the walkway stair) | exact (2 risers @ 5'-0") |
| 2 | **2** | ea | Steel stair stringer, galvanized, pre-formed to the riser / tread | exact |

## Porch cover

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 2 | **2** | ea | 6x6x10 Douglas fir rough post (cover) — Main cover | exact (2 posts on the beam line, field-cut) |
| 1 | **1** | ea | 4x10x14 Douglas fir — cover beam — Main cover | +0 (4x10 x 12.8' at the posts, one piece) |
| 1 | **1** | ea | 2x8x16 #1 SYP — cover ledger on the house — Main cover | exact |
| 10 | **11** | ea | 2x8x14 #1 SYP — cover rafters @ 16" OC — Main cover | +1 cull (10 rafters, 13.2' with the overhang) |
| 3 | **4** | ea | 2x8x16 #1 SYP — cover fascia / sub-fascia — Main cover | +1 (38 LF) |
| 6 | **6** | sheet | 7/16" OSB roof sheathing 4x8 — Main cover | 170 SF + 10% cuts |
| 170 | **178** | SF | Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 13.2' (Black, PVDF), 10 panels — Main cover | 10 panels x 13.2' + 5% (one-piece eave to headwall, no end laps) — mechanical seam priced at the snap-lock rate; confirm with the metal supplier |
| 1 | **1** | roll | High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Main cover | 170 SF — full coverage under metal (high-temp) |
| 2 | **3** | ea | Standing seam eave / drip trim 10', color-matched — Main cover | 12 LF low eave + 1 |
| 3 | **4** | ea | Standing seam gable / rake trim 10', color-matched — Main cover | two rakes x 13.2' + 1 |
| 2 | **3** | ea | Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Main cover | 12 LF at the house — counterflash into the siding / brick |
| 3 | **3** | ea | Foam + metal closure strips at eave and headwall, 10' — Main cover | eave + headwall closures |
| 1 | **1** | box | Panel clips + pancake-head screws, box of 250 — Main cover | 90 clips @ 18" OC on each seam |
| 1 | **1** | box | Color-matched #10 woodgrip trim screws w/ washers, 250 — Main cover | trims, closures, snow bar |
| 3 | **3** | ea | Butyl tape / polyurethane sealant, per roll or tube — Main cover | headwall, eave, rake laps |
| 12 | **14** | LF | Snow retention — clamp-on bar across the low eave over the deck (Black) — Main cover | 12' — metal sheds 30 psf snow onto the deck and stair below; one bar 12-18" above the eave |
| 5 | **5** | panel | Porch ceiling — HardieSoffit 4' x 10' cedarmill ColorPlus Arctic White (non-vented; 1 in 3 vented if the roof is closed) — Main cover | 170 SF + 10% cuts |

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 3 | **3** | lb | Stainless ring-shank siding nails 2", per lb — Main cover | stainless ring-shank, 6" OC at every rafter |
| 4 | **5** | ea | Cover fascia + rakes — HardieTrim 4/4 x 7-1/4 x 12' rustic ColorPlus Arctic White over the 2x8 sub-fascia — Main cover | +1 (38 LF: front 12' + two rakes 13.2') |
| 1 | **1** | box | Color-matched trim screws for HardieTrim, 100 ct (Timber Bark) — Main cover | color-matched trim screws |
| 2 | **2** | ea | Simpson ABA66RZ post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Main cover | exact |
| 2 | **2** | ea | Simpson BC46Z post cap, 6x6 post to the 4x beam — sized for a rough 6x6 (R-series), confirm SKU — Main cover | exact |
| 10 | **10** | ea | Simpson H2.5AZ rafter tie — Main cover | one per rafter at the beam |
| 19 | **19** | ea | FastenMaster LedgerLOK 6" — cover ledger, 2 rows staggered 16" OC — Main cover | 12 LF x 2 rows @ 16" |
| 2 | **2** | ea | 6x6x10 Douglas fir rough post (cover) — Walkway cover | exact (2 posts on the beam line, field-cut) |
| 2 | **2** | ea | 2x8x16 #1 SYP — cover beam (2 ply) — Walkway cover | +0 ((2)2x8 x 6.3' at the posts) |
| 1 | **1** | ea | 2x8x16 #1 SYP — cover ledger on the house — Walkway cover | exact |
| 5 | **6** | ea | 2x8x8 #1 SYP — cover rafters @ 16" OC — Walkway cover | +1 cull (5 rafters, 4.9' with the overhang) |
| 2 | **3** | ea | 2x8x16 #1 SYP — cover fascia / sub-fascia — Walkway cover | +1 (16 LF) |
| 2 | **2** | sheet | 7/16" OSB roof sheathing 4x8 — Walkway cover | 35 SF + 10% cuts |
| 35 | **37** | SF | Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 4.9' (Black, PVDF), 5 panels — Walkway cover | 5 panels x 4.9' + 5% (one-piece eave to headwall, no end laps) — mechanical seam priced at the snap-lock rate; confirm with the metal supplier |
| 1 | **1** | roll | High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Walkway cover | 35 SF — full coverage under metal (high-temp) |
| 1 | **2** | ea | Standing seam eave / drip trim 10', color-matched — Walkway cover | 6 LF low eave + 1 |
| 1 | **2** | ea | Standing seam gable / rake trim 10', color-matched — Walkway cover | two rakes x 4.9' + 1 |
| 1 | **2** | ea | Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Walkway cover | 6 LF at the house — counterflash into the siding / brick |
| 2 | **2** | ea | Foam + metal closure strips at eave and headwall, 10' — Walkway cover | eave + headwall closures |
| 1 | **1** | box | Panel clips + pancake-head screws, box of 250 — Walkway cover | 20 clips @ 18" OC on each seam |
| 1 | **1** | box | Color-matched #10 woodgrip trim screws w/ washers, 250 — Walkway cover | trims, closures, snow bar |
| 3 | **3** | ea | Butyl tape / polyurethane sealant, per roll or tube — Walkway cover | headwall, eave, rake laps |
| 6 | **8** | LF | Snow retention — clamp-on bar across the low eave over the deck (Black) — Walkway cover | 6' — metal sheds 30 psf snow onto the deck and stair below; one bar 12-18" above the eave |

| NET | ORDER | UNIT | ITEM | WHY |
|---|---|---|---|---|
| 1 | **1** | panel | Porch ceiling — HardieSoffit 4' x 10' cedarmill ColorPlus Arctic White (non-vented; 1 in 3 vented if the roof is closed) — Walkway cover | 35 SF + 10% cuts |
| 1 | **1** | lb | Stainless ring-shank siding nails 2", per lb — Walkway cover | stainless ring-shank, 6" OC at every rafter |
| 2 | **3** | ea | Cover fascia + rakes — HardieTrim 4/4 x 7-1/4 x 12' rustic ColorPlus Arctic White over the 2x8 sub-fascia — Walkway cover | +1 (16 LF: front 6' + two rakes 4.9') |
| 1 | **1** | box | Color-matched trim screws for HardieTrim, 100 ct (Timber Bark) — Walkway cover | color-matched trim screws |
| 2 | **2** | ea | Simpson ABA66RZ post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Walkway cover | exact |
| 2 | **2** | ea | Simpson AC6RZ post cap, rough 6x6 to the (2)2x8 beam — Walkway cover | exact |
| 5 | **5** | ea | Simpson H2.5AZ rafter tie — Walkway cover | one per rafter at the beam |
| 10 | **10** | ea | FastenMaster LedgerLOK 6" — cover ledger, 2 rows staggered 16" OC — Walkway cover | 6 LF x 2 rows @ 16" |

## FASTENER & CONNECTOR SCHEDULE

| | |
|---|---|
| Ledger | 600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood |
| Field | CONCEALoc at every joist in every gap (437) |
| Face screws | Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled · stair treads 2 per stringer per board |
| Steel frame | 600S162-54 C-joists @ 12" OC · single-ply 600T track rims · boxed 600S162-54 beams · all G90 · clip angles with #12 self-drilling screws (4 per leg) · cut ends cold-galvanized · no joist tape needed |

## CUT LIST

| MEMBER | STOCK | LENGTH | QTY | NOTE |
|---|---|---|---|---|
| Field joists (W) | 600S162-54 | 3'-3 1/8" | 13 | @ 12" OC, crown up |
| Picture-frame joists (W) | 600S162-54 | 3'-3 1/8" | 2 | centre 5 9/16" from each frame face |
| Side rim plies (W) | 600S162-54 | 3'-3 1/8" | 4 | 2-ply, laminated on the ground with RSS 2 rows @ 12" staggered |
| Ledger (W) | 600S162-54 | 14'-8" | 1 | top at deck height less board thickness; membrane behind, flashing over |
| Blocking (W) | 600S162-54 | 10 3/8" | 14 | 1 row at 1'-10" from the wall |
| Field joists (M) | 600S162-54 | 11'-7 1/8" | 11 | @ 12" OC, crown up |
| Picture-frame joists (M) | 600S162-54 | 11'-7 1/8" | 2 | centre 5 9/16" from each frame face |

| MEMBER | STOCK | LENGTH | QTY | NOTE |
|---|---|---|---|---|
| Divider doublers (M) | 600S162-54 | 11'-7 1/8" | 1 | sister a joist on the divider line so both board edges bear |
| Side rim plies (M) | 600S162-54 | 11'-7 1/8" | 4 | 2-ply, laminated on the ground with RSS 2 rows @ 12" staggered |
| Ledger (M) | 600S162-54 | 12'-0" | 1 | top at deck height less board thickness; membrane behind, flashing over |
| Blocking (M) | 600S162-54 | 10 3/8" | 12 | 1 row at 6'-0" from the wall |
| Field boards panel 1 | Landmark American Walnut | 14'-0" | 6 | one board wall to border, no joints |
| Field boards panel 1 (rip) | Landmark American Walnut | 14'-0" | 1 | one board wall to border, no joints; ripped to fit the border / the house |
| Field boards panel 2 | Landmark American Walnut | 11'-3" | 18 | one board wall to border, no joints |
| Field boards panel 2 | Landmark American Walnut | 11'-4" | 6 | one board wall to border, no joints |
| Field boards panel 2 (rip) | Landmark American Walnut | 11'-4" | 1 | one board wall to border, no joints; ripped to fit the border / the house |
| Stair stringers (main deck side, toward the walkway) | steel stringer | 2'-7" | 6 | 2 risers @ 7.500" · 1 treads @ 11.66" · stringers 12" OC |
| FLUSH boxed 600S162-54 W beam | 600S162-54 | 14'-1 1/2" | 2 | y 3'-6 3/8"; posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" |
| 6x6 deck posts / cover posts under FLUSH boxed 600S162-54 W | 6x6 DF rough | 6" | 4 | field-measure each |
| FLUSH boxed 600S162-54 M beam | 600S162-54 | 11'-5 1/2" | 2 | y 11'-10 3/8"; posts at 15'-2" / 20'-8" / 26'-2" |
| 6x6 deck posts / cover posts under FLUSH boxed 600S162-54 M | 6x6 DF rough | 6" | 3 | field-measure each |
| Border 26'-8",0" | Landmark American Walnut square | 12'-1" | 1 | mitre at the corners; dividers on the doubled joist |
| Border 14'-8",12'-0" | Landmark American Walnut square | 12'-1" | 1 | mitre at the corners; dividers on the doubled joist |
| Border 14'-8",3'-8" | Landmark American Walnut square | 8'-4 15/16" | 1 | mitre at the corners; dividers on the doubled joist |
| Border 0",3'-8" | Landmark American Walnut square | 14'-9" | 1 | mitre at the corners; dividers on the doubled joist |
| Border 0",0" | Landmark American Walnut square | 3'-9" | 1 | mitre at the corners; dividers on the doubled joist |
| Breaker dw/m | Landmark American Walnut square | 12'-1" | 1 | mitre at the corners; dividers on the doubled joist |
| Stair treads (main deck side, toward the walkway) | Landmark American Walnut square | 5'-0" | 2 | 2 boards per tread, nosing 3/4" over the riser |
| Stair risers (main deck side, toward the walkway) | riser board | 5'-0" | 2 | rip to 7.50" less tread thickness |

| MEMBER | STOCK | LENGTH | QTY | NOTE |
|---|---|---|---|---|
| Left fascia (w) | fascia | 3'-8 1/2" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (w) | fascia | 3'-8 1/2" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (w) | fascia | 12'-0" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (w) | fascia | 2'-9" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (m) | fascia | 12'-0" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Left fascia (m) | fascia | 1/2" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (m) | fascia | 12'-0" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Right fascia (m) | fascia | 1/2" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (m) | fascia | 12'-0" | 1 | flush under the board nose; full board at the front corners, short piece at the house |
| Front fascia (m) | fascia | 1" | 1 | flush under the board nose; full board at the front corners, short piece at the house |

# G-001 General notes · design criteria

## G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0

### PROJECT

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel, TimberTech Landmark American Walnut)
281 S Jefferson Ave, Louisville CO 80027
197.8 SF · 1'-3" above grade · 2 zones

### DESIGN CRITERIA

| | |
|---|---|
| **Deck live load** | 40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf |
| **Ground snow** | 30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; ground snow per the ASCE hazard tool (Louisville criteria) — 30 psf used here (city sheet, non-reducible); Boulder County layer shows 40 psf: engineer confirms |
| **Dead load** | 10 psf deck · 15 psf porch cover (standing seam steel on sheathing, Hardie soffit; 12 psf actual, 15 used) carried on the deck posts |
| **Roof snow (cover)** | 30 psf flat-roof design snow, sliding snow off the house roof + drift at the house wall per ASCE 7 Ch. 7 |
| **Design total** | 50 psf on the deck · cover posts add roof snow + dead to the deck frame and footings (engineered) |
| **Wind** | 145 mph Vult (3-second gust), Exposure C (Louisville 2024 design criteria), Risk Cat. II |
| **Frost depth** | 3'-0" (footings bear below) |
| **Soil bearing** | 1,500 psf presumptive (IRC R401.4.1) — no soils report |
| **Seismic** | B |
| **Wildfire (WUI)** | not in a WUI zone |
| **Codes** | 2024 IRC / IBC as adopted and amended by the City of Louisville (Ord. 1907, Series 2025); IRC R507 decks with LMC 15.05.140; IBC Appendix I patio covers; ASCE 7-22 loads; Louisville Fire Hardening provisions; cold-formed steel frame per AISI S240, engineered |
| **Engineering** | Stamped structural set by a Colorado PE required — the stamped set governs |

### STRUCTURE

| | |
|---|---|
| **Footings** | DP-50/50 |
| **Posts** | 3 x 6x6 Douglas fir rough deck posts |
| **Beams** | FLUSH boxed 600S162-54 W: 14'-1 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 471 lb); FLUSH boxed 600S162-54 M: 11'-5 1/2", 3 posts at 15'-2" / 20'-8" / 26'-2" (worst 1,613 |
| **Joists** | CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) @ 12" OC |
| **Rims** | single-ply 600T track rims (G90) and 600T ledger track on a steel angle |
| **Ledger** | 600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood |
| **Decking** | TimberTech Landmark American Walnut — 25 rows parallel to the house, picture frame & dividers in Landmark American Walnut, 3/16" gaps, CONCEALoc |
| **Rail** | none |

### SHEET INDEX


| | |
|---|---|
| G-001 | General notes · design criteria |
| C-100 | Site plan |
| A-001 | Isometric · renders |
| A-002 | Isometric — assembled, labeled |
| A-003 | Isometric — exploded, labeled |
| S-100 | Foundation plan |
| S-101 | Framing plan |
| S-102 | Structure isometric, labeled |
| A-101 | Decking plan |
| A-202 | Elevations |
| A-301 | Section + front elevation |
| A-401 | Roof covers — plan + section |
| A-402 | Roof covers — exploded isometric |
| D-501 | Details |
| D-502 | Connection details — steel frame |
| D-503 | Connection details — roof covers |
| T-400 | Build steps (build set) |
| T-700 | Order + schedules (takeoff) |


### SITE

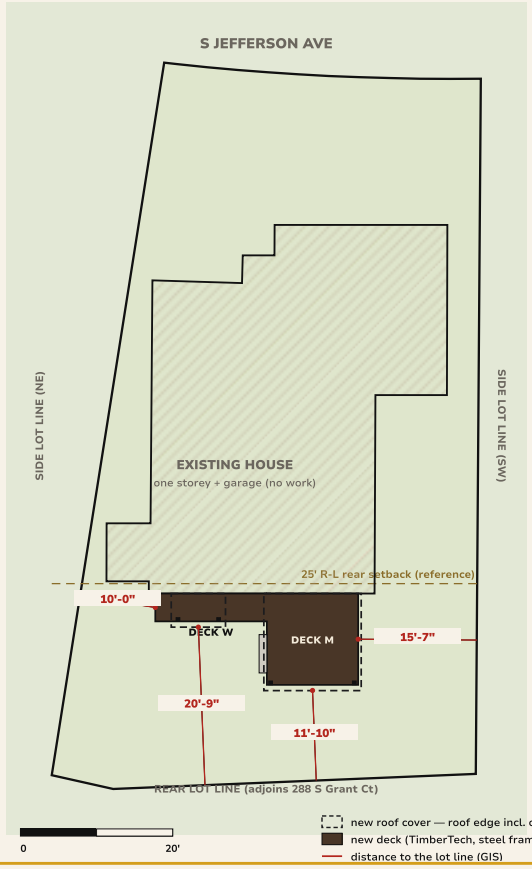
Rear / front lot line: deck edge 12.0' from the line, setback per the PUD...
Side lot line: deck edge 9.8' from the line, setback per the PUD —...

# C-100 SITE PLAN

Parcel, existing house, new decks and roof covers, distances to the lot lines, coverage

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0







## PARCEL

281 S Jefferson Ave, Louisville CO 80027
Parcel 157517119033 · Account R0092293
Lot 33, Block 10, Dutch Creek (plat rec. 00492001, 1982)
Section 17, T1S, R69W
Zoning: R-L within the Dutch Creek PUD (approved 12/15/1981)
Lot 4,678 SF · flood zone X
House 1982 one-storey ranch: 920 SF main floor + 400 SF garage (assessor)

## LOT COVERAGE

| | |
|---|---|
| **Lot area** | 4,678 SF |
| **House + garage (assessor footprint)** | 1,320 SF |
| **Existing covers to the posts (2024 aerial, est.)** | ~142 SF (main ~12' x 10'-4", walkway ~5'-4" x 3'-5") |
| **Existing coverage** | ~1,462 SF · 31.3% |
| **Main cover, new (to the posts) 12'-0" x 11'-9"** | 141 SF |
| **Walkway cover, new (to the posts) 5'-4" x 3'-5"** | 18 SF |
| **Proposed coverage** | ~1,479 SF · 31.6% (+~17 SF) |
| **Uncovered decks** | not counted (under 30" above grade) |

## SETBACKS

Decks are 30" or less above grade — not held to the principal setbacks (city deck handout)
Roof covers: part of the principal structure — they replace the two existing covers in place; the 2024 aerial shows the existing main roof ~1'-5" shallower: field-verify the existing post line
R-L reference setbacks (LMC 17.12.040) — the Dutch Creek PUD plan governs; Planning to confirm
Measured (red): to the nearest lot line from the new work — GIS, not survey

## SITE WORK

· Remove the two decks, the two roof covers and the box step; haul off
· Diamond Pier foundations (no excavation spoils), 6-mil poly + 2" gravel under the decks
· Stair lands on grade (no pad)
· Roof drainage: covers shed to grade at the low eave, away from the house
· Call 811 before the piers; protect the lawn and the owner's blinds

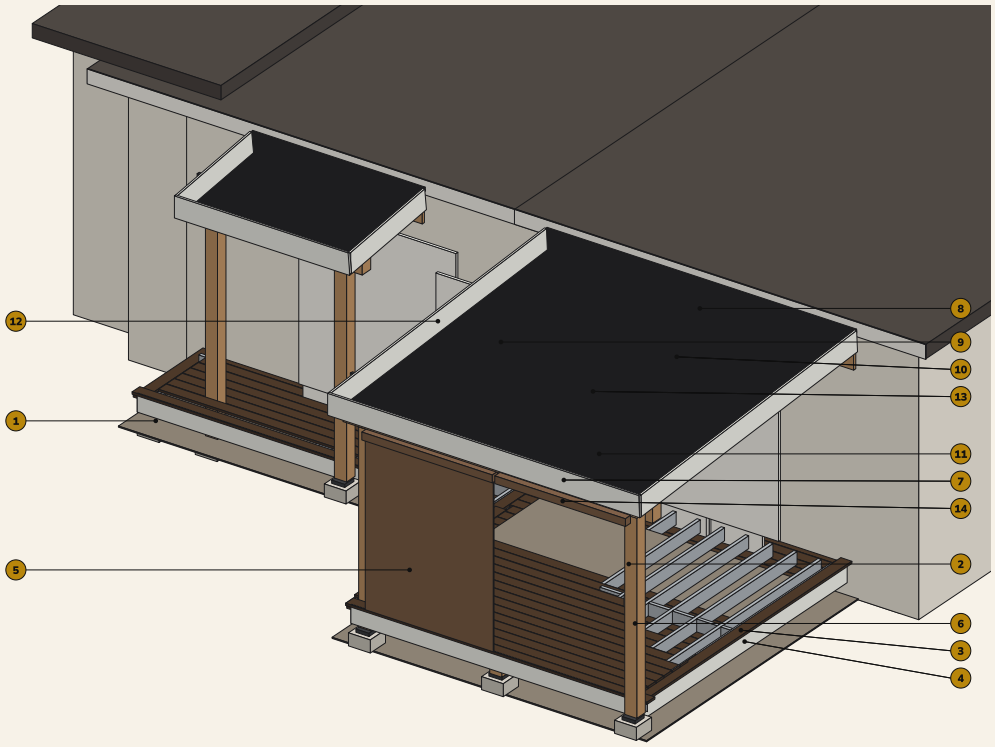
Boulder County GIS parcel + 2024 Pictometry roof outline, Louisville GIS zoning / PUD layers. GIS lines are not a survey — an ILC governs if Planning requires one.

# A-002 ISOMETRIC — ASSEMBLED, LABELED

Everything you see called out: piers, decking, picture frame, skirt, stair, roof covers, finishes

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0




**KEY**

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. TimberTech Landmark American Walnut field decking, hidden fasteners
3. Picture frame + breaker boards — TimberTech Landmark American Walnut
4. HardieTrim skirt Arctic White
5. Stair — steel stringers, Landmark treads, closed risers, lands on grade
6. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)
7. 4x10 cover beam, Douglas fir, BC46Z cap
8. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
9. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
10. Black standing seam, 24 ga, mechanically seamed
11. Snow retention bar, clamped to the seams
12. HardieTrim fascia + rakes Arctic White
13. HardieSoffit cedarmill ceiling Arctic White
14. Existing roll-down blinds (stay)

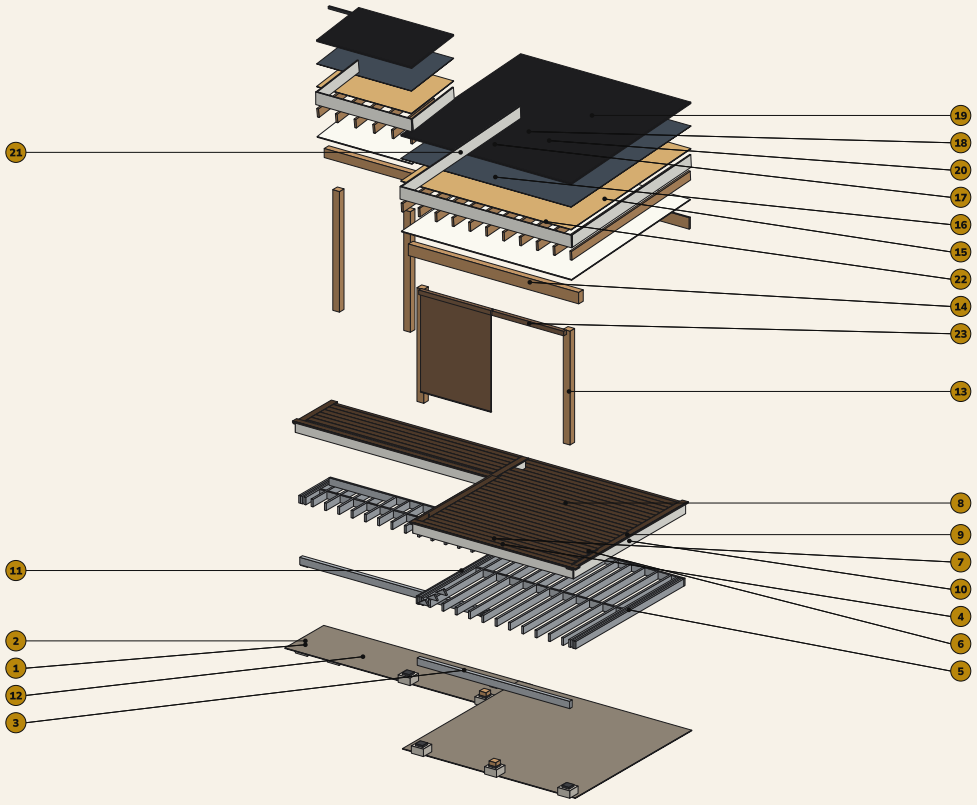
House shown for context only (no work). The CEMCO steel deck frame under the decking is called out on S-102 and A-003; cover framing is wood: 2x8 rafters, main cover 4x10 beam, walkway cover (2)2x8 beam, 6x6 Douglas fir posts.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

# A-003 Isometric — exploded, labeled






KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir rough deck post on ABA66RZ, steel saddle to the boxed beam
3. Boxed CEMCO 600S162-54 beam (2 C toe-to-toe, G90), flush
4. CEMCO 600S162-54 C-joist, G90, @ 12" OC — clip angles each end
5. CEMCO 600T track rim, G90 (caps the joist ends)
6. Steel ledger track + 3x3 ledger angle on the house rim, flashed
7. Steel blocking at the breaker / picture-frame joists
8. TimberTech Landmark American Walnut field decking, hidden fasteners
9. Picture frame + breaker boards — TimberTech Landmark American Walnut
10. HardieTrim skirt Arctic White
11. Stair — steel stringers, Landmark treads, closed risers, lands on grade
12. Gravel over 6-mil poly under the decks
13. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)
14. 4x10 cover beam, Douglas fir, BC46Z cap
15. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
16. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
17. 7/16" OSB roof sheathing
18. High-temp self-adhered underlayment
19. Black standing seam, 24 ga, mechanically seamed
20. Snow retention bar, clamped to the seams
21. HardieTrim fascia + rakes Arctic White
22. HardieSoffit cedarmill ceiling Arctic White
23. Existing roll-down blinds (stay)

Layers lifted in build order. Roof covers pulled apart: posts, beam + ledger, rafters, soffit, sheathing, underlayment, standing seam, snow bar, Hardie trim.

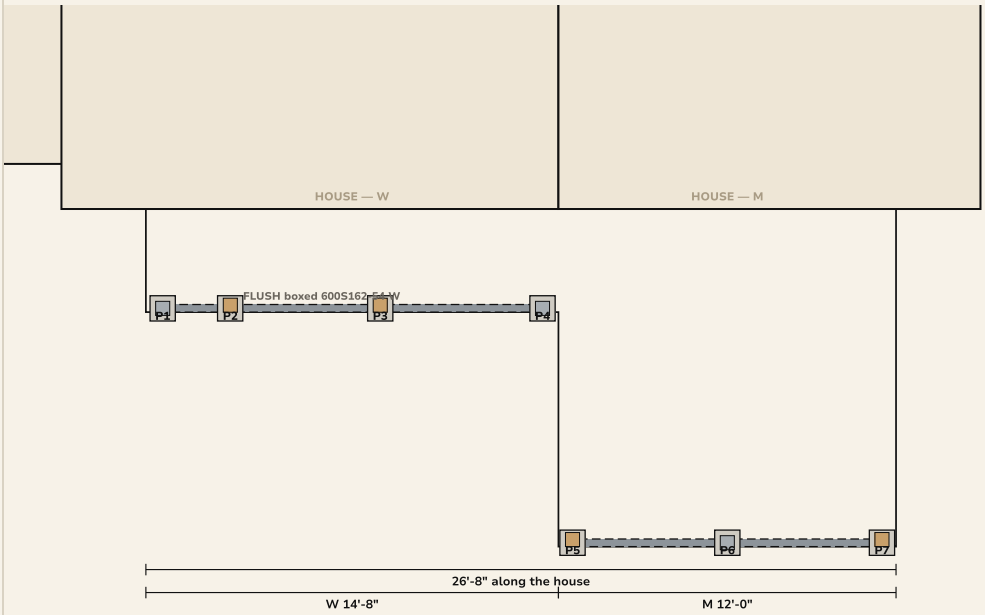
S-100

# FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0




## FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

| ID | X | Y | ZONE | LOAD |
|---|---|---|---|---|
| P1 | 7 3/16" | 3'-6 3/8" | W | 130 lb |
| P2 | 3'-0" | 3'-6 3/8" | W +cover | 328 lb +roof |
| P3 | 8'-4" | 3'-6 3/8" | W +cover | 471 lb +roof |
| P4 | 14'-1 3/16" | 3'-6 3/8" | W | 270 lb |
| P5 | 15'-2" | 11'-10 3/8" | M +cover | 874 lb +roof |
| P6 | 20'-8" | 11'-10 3/8" | M | 1,613 lb |
| P7 | 26'-2" | 11'-10 3/8" | M +cover | 873 lb +roof |

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 1,613 lb

Deck-only piers: 6x6 Douglas fir rough post on ABA66RZ, steel saddle to the bea
+cover: the 6x6 Douglas fir rough roof-cover post stands on this pier (ABA66RZ) and carries the deck beam on a steel side bracket (D-502 / 4); roof load adds — the engineer sizes the pier.
Beams: boxed CEMCO 600S162-54, flush (D-502 / 2).

Plan · not to scale · house at the top · dimensions govern

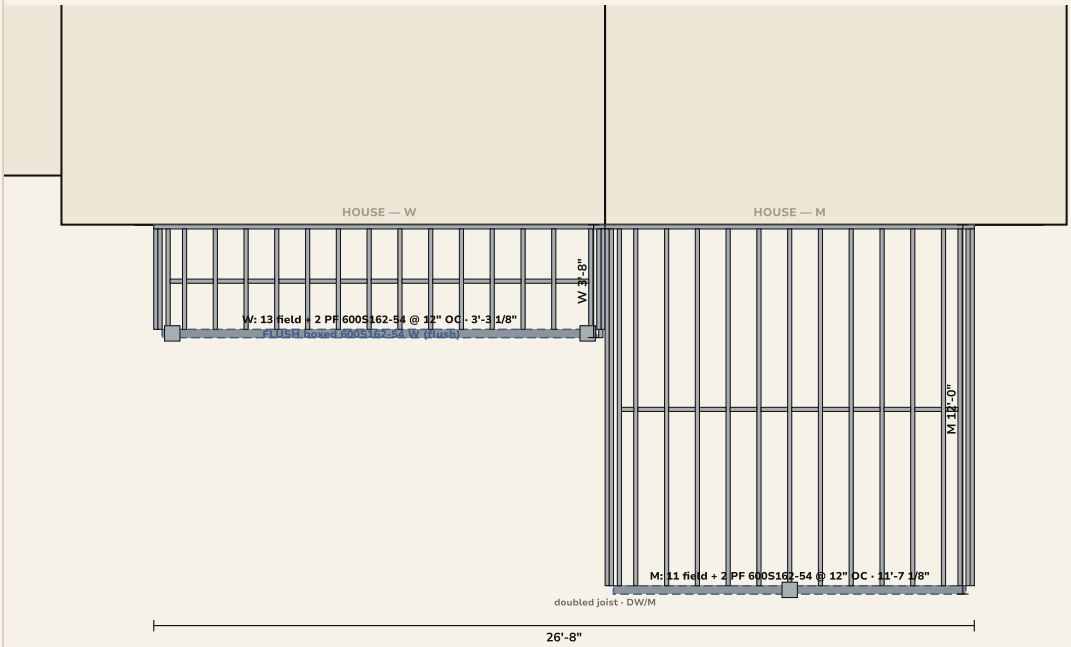
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

# S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0




## FRAMING NOTES

CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) @ 12" OC

single-ply 600T track rims (G90) and 600T ledger track on a steel angle

600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer

blocking one row over the beam, staggered

FLUSH boxed 600S162-54 W: 14'-1 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 471 lb); FLUSH boxed

Connectors:

Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam: 56

Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2

Cold-galvanizing spray (cut ends, field drilling): 2

Plan · not to scale · house at the top · dimensions govern

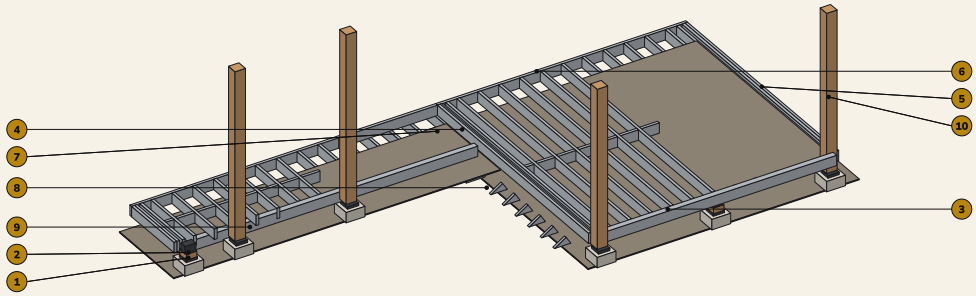
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

# S-102 STRUCTURE ISOMETRIC — STEEL FRAME, LABELED

Piers, stubs, cover posts at the shared piers, boxed beams, C-joists, track rims, ledger track, blocking

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0




**KEY**

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir rough deck post on ABA66RZ, steel saddle to the boxed beam
3. Boxed CEMCO 600S162-54 beam (2 C toe-to-toe, G90), flush
4. CEMCO 600S162-54 C-joist, G90, @ 12" OC — clip angles each end
5. CEMCO 600T track rim, G90 (caps the joist ends)
6. Steel ledger track + 3x3 ledger angle on the house rim, flashed
7. Steel blocking at the breaker / picture-frame joists
8. Stair — steel stringers, Landmark treads, closed risers, lands on grade
9. Gravel over 6-mil poly under the decks
10. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)

Decking, covers and house removed to show the frame. Every joist end clips to the track, ledger track or boxed beam; the engineer's clip schedule governs.

# A-101 Decking plan

**A-101** **DECKING PLAN**

Every board, border and divider as it lays; rail posts shown for the pockets

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0




## DECKING

Field: TimberTech Landmark American Walnut, 5.5" x 1", 3/16" gaps

Border & dividers: Landmark American Walnut

Boards run parallel to the house; edge overhang 1" over the fascia

CONCEALoc at every joist in every gap (437)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per panel: panel 1 (0"–14'-8") 7 rows · panel 2 (14'-8"–26'-8") 25 rows · 2 ripped pieces · 1 breaker — every board one piece wall to border, no butt

Plan · not to scale · house at the top · dimensions govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

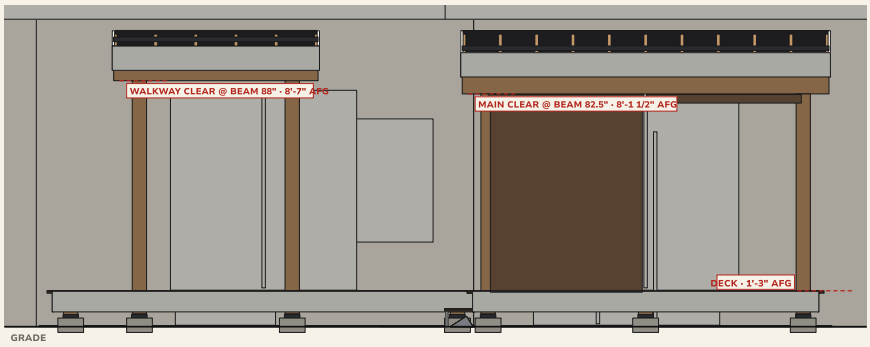
# A-202 ELEVATIONS

Front from the yard and both ends, drawn from the model — deck and roof-cover heights

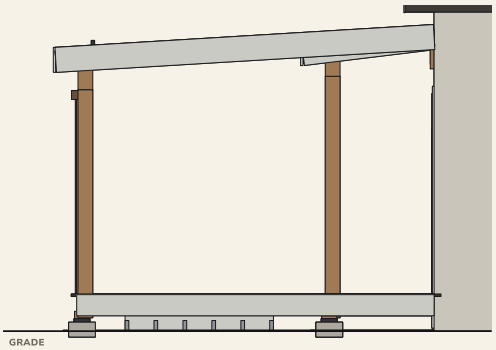
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0

## FRONT ELEVATION — from the yard



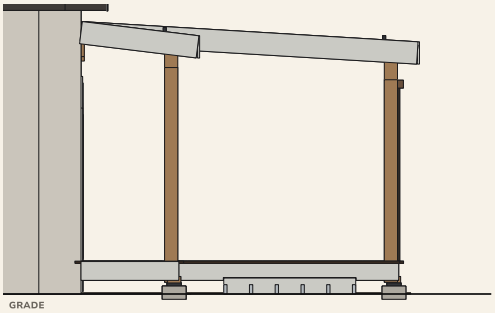

## RIGHT END ELEVATION




## NOTES

- Deck surface 1'-3" above grade (walkway low point 9")
- Heights AFG = above finished grade at the beam line; field-verify grade before setting the piers
- Clear heights are to the underside of the beam / ledger above the deck surface
- Roof covers: black standing seam on 2x8 rafters, Hardie fascia / rakes / soffit; slope to the yard
- Deck frame: CEMCO cold-formed steel; HardieTrim skirt covers the track rims
- No guard required: every walking surface is under 30" above grade (IRC R312.1)

## LEFT END ELEVATION






Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

# A-301 Section + front elevation

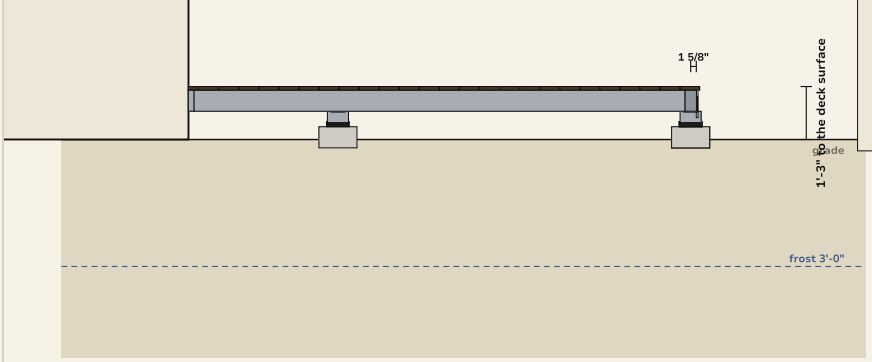
## A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

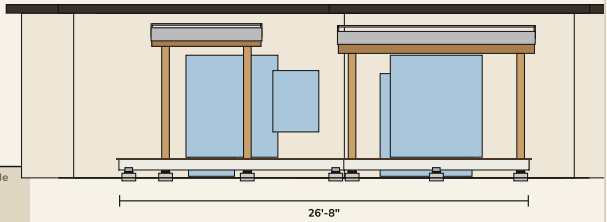
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0

**SECTION through zone M at 6'-0" — looking toward the left end**




**FRONT ELEVATION — from the yard**




### SECTION NOTES

Deck surface 1'-3" above grade

FLUSH boxed 600S162-54 W: 14'-1 1/2", 4 posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16" (worst 471 lb); FLUSH boxed 600S162-54 M: 11'-5 1/2", 3 posts at 15'-2" / 20'-8" / 26'-2"

Posts 3 x 6x6 Douglas fir rough deck posts on steel beam saddles + 4 6x6 Douglas fir rough cover posts carrying the deck beam on DP-50/50

600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

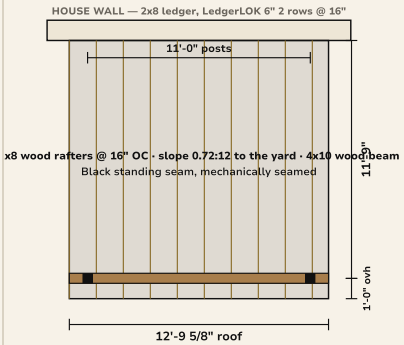
# A-401 ROOF COVERS — PLAN + SECTION

Lean-to covers: ledger, rafters, beam, posts on the piers, clear heights, pitch, standing seam, Hardie, blinds

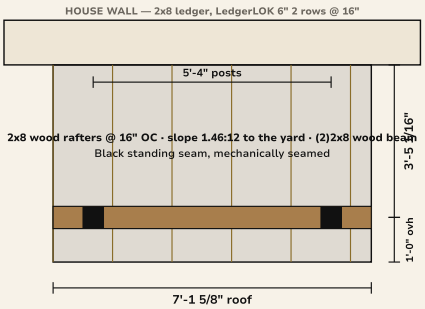
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0

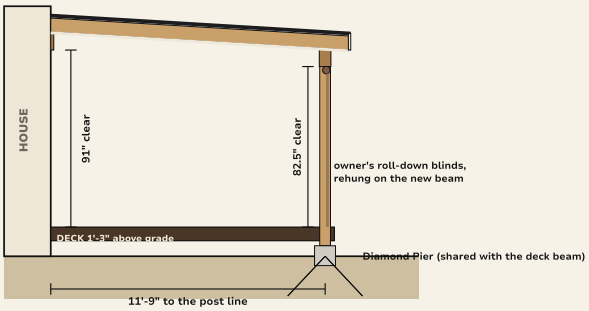
## MAIN COVER — PLAN




## WALKWAY COVER — PLAN

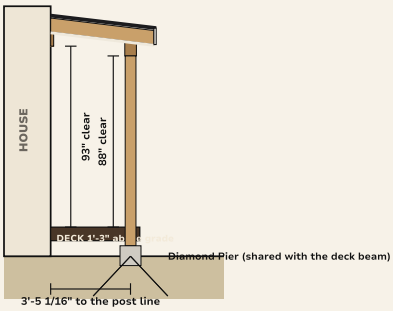



## MAIN COVER — SECTION




Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 0.72:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · post caps per D-503
6x6 Douglas fir rough posts on ABA66RZ bases on the Diamond Pier bolt · BC46Z cap to the 4x10 Douglas fir beam · the engineer's uplift hold-down governs

## WALKWAY COVER — SECTION




Standing seam 24 ga, 16" panels, 1-1/2" mechanical (double-lock) seam — rated for 1.46:12 · high-temp ice & water underlayment over 7/16" OSB
Headwall flashing counterflashed into the siding · color-matched eave / rake trim · snow bar across the low eave
HardieSoffit cedarmill ceiling · HardieTrim fascia + rakes (Arctic White) · H2.5AZ at every rafter · post caps per D-503
6x6 Douglas fir rough posts on ABA66RZ bases on the Diamond Pier bolt · AC6Z cap to the (2)2x8 beam · the engineer's uplift hold-down governs

Plans and sections not to a common scale · dimensions govern · the stamped set governs

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

# A-402 Roof covers — exploded isometric, labeled

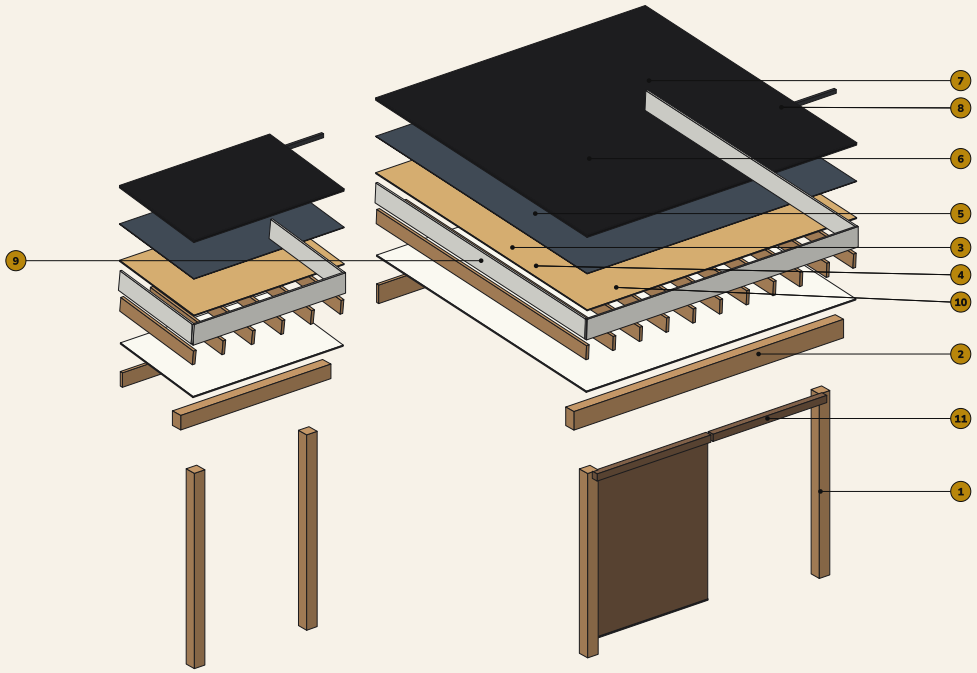
A-402

## ROOF COVERS — EXPLODED ISOMETRIC, LABELED

Both lean-to covers pulled apart layer by layer

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0




**KEY**

1. 6x6 Douglas fir rough cover post on ABA66RZ (shared Diamond Pier)
2. 4x10 cover beam, Douglas fir, BC46Z cap
3. 2x8 cover ledger, wood, LedgerLOK 6" 2 rows, headwall flashed
4. 2x8 rafters @ 16" OC, wood, H2.5AZ at the beam
5. 7/16" OSB roof sheathing
6. High-temp self-adhered underlayment
7. Black standing seam, 24 ga, mechanically seamed
8. Snow retention bar, clamped to the seams
9. HardieTrim fascia + rakes Arctic White
10. HardieSoffit cedarmill ceiling Arctic White
11. Existing roll-down blinds (stay)

Lifts are for clarity only. Order of work: posts on ABA66RZ bases, beams on their post caps, ledger on LedgerLOK, rafters with H2.5AZ, soffit, OSB, underlayment, panels, trims, snow bar.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

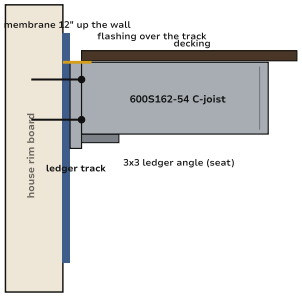
# D-501 DETAILS — STEEL FRAME

Steel ledger · joist clips · boxed beam on the shared pier · Diamond Pier · deck edge · under the deck

281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

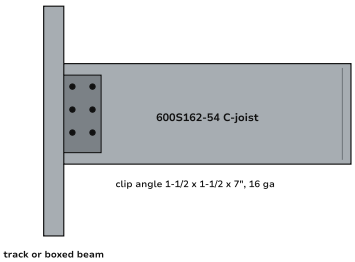
GS Exterior Experts · October 16, 2026 · Rev 0

## 1 STEEL LEDGER AT THE HOUSE



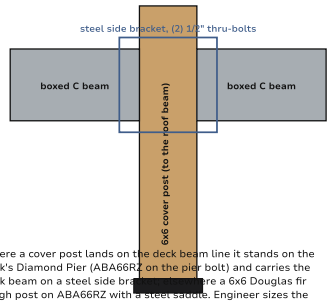

600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood

## 2 JOIST TO TRACK / BEAM



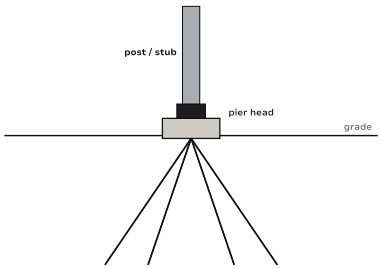

#12-14 self-drilling hex-washer screws, 4 per leg minimum (clip to joist web, clip to track / beam web) — the engineer's clip schedule governs. Cut ends and field holes cold-galvanized.

## 3 BOXED BEAM · SHARED PIER



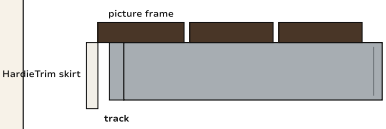

Where a cover post lands on the deck beam line it stands on the deck's Diamond Pier (ABA66RZ on the pier bolt) and carries the deck beam on a steel side bracket; above it is a 6x6 Douglas fir rough post on ABA66RZ with a steel saddle. Engineer sizes the pier for roof + deck and the uplift hold-down.

## 4 DIAMOND PIER




DP-50/50 (4 pins) — no excavation; frost-protected by the pins (submit the evaluation report). 811 locate first.

## 5 DECK EDGE · HARDIE SKIRT




TimberTech Landmark American Walnut: hidden fasteners in the field, the picture frame face-fastened with color-matched plugs — every screw the maker's steel-frame screw for 14–18 ga. HardieTrim skirt on furring screwed to the track.

## 6 UNDER THE DECK · LOW END




2" gravel over 6-mil poly, sloped away from the house

Keep 2" min. between the frame and the gravel.

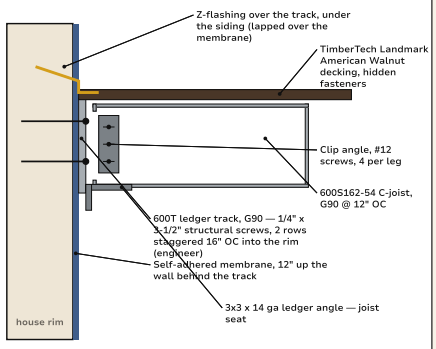
# D-502 CONNECTION DETAILS — CEMCO STEEL DECK FRAME

Ledger track · joist clips · boxed beam · post on the pier · shared pier · deck edge · stair

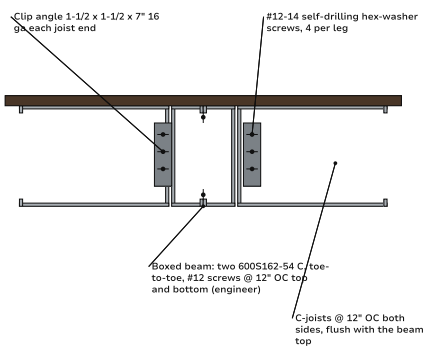
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0

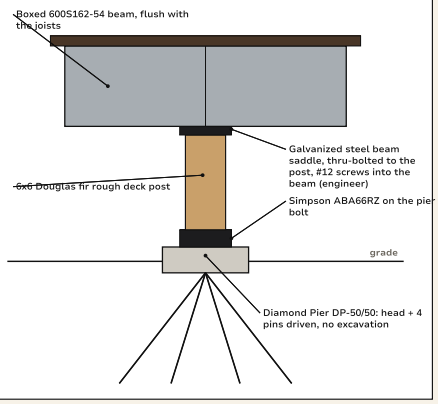
## 1 LEDGER TRACK AT THE HOUSE — SECTION



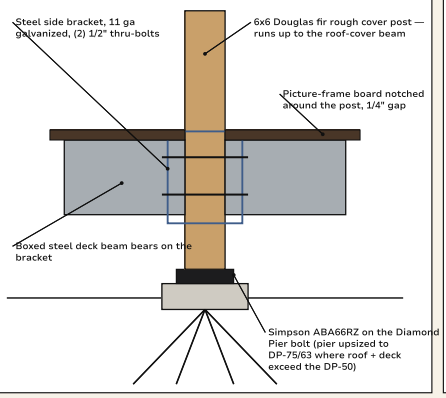

## 2 JOIST TO BOXED BEAM — SECTION



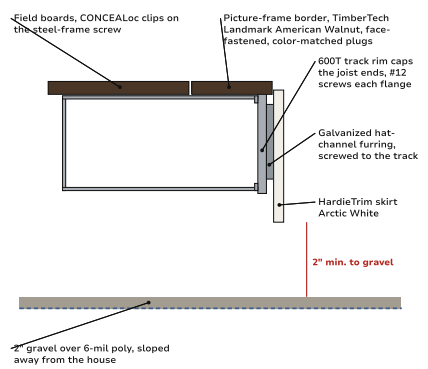

## 3 BEAM ON 6x6 POST + DIAMOND PIER



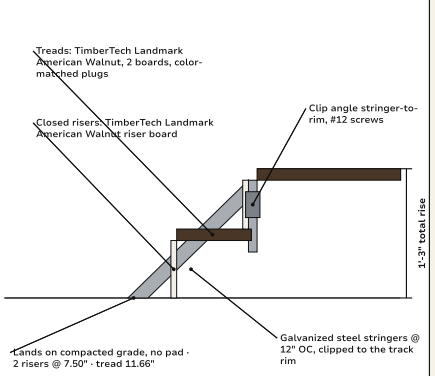

## 4 SHARED PIER — COVER POST + DECK BEAM




## 5 DECK EDGE — TRACK RIM, BORDER, HARDIE SKIRT




## 6 STAIR OFF THE MAIN DECK — SECTION



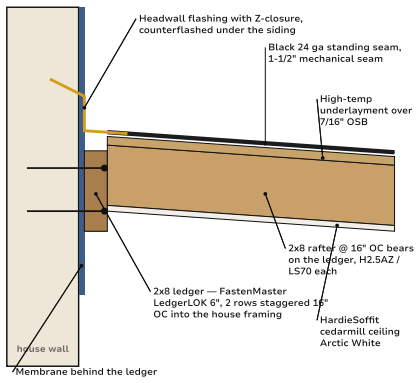
# D-503 CONNECTION DETAILS — ROOF COVERS (WOOD FRAMING)

Ledger + headwall · eave at the beam · post base · rake · standing seam + clip · blinds

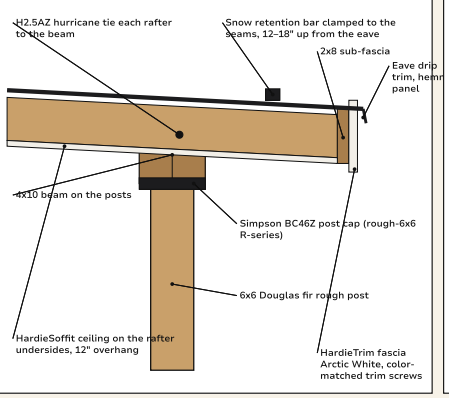
281 S Jefferson Ave — two decks + two lean-to covers (CEMCO steel,...

GS Exterior Experts · October 16, 2026 · Rev 0

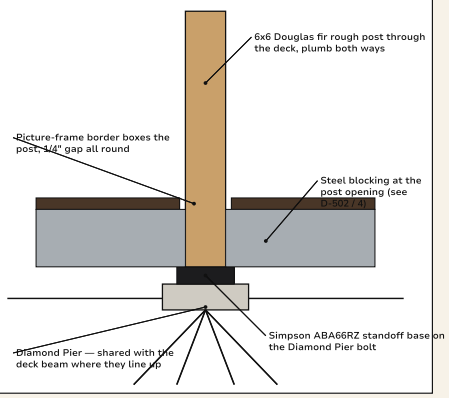
## 1 COVER LEDGER + HEADWALL — SECTION



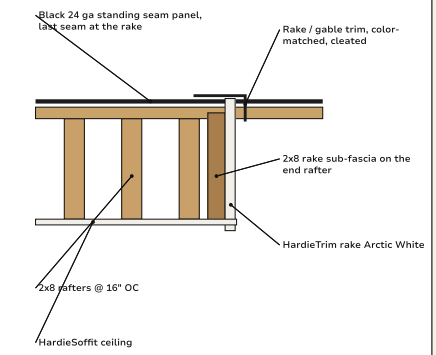

## 2 EAVE AT THE BEAM — SECTION



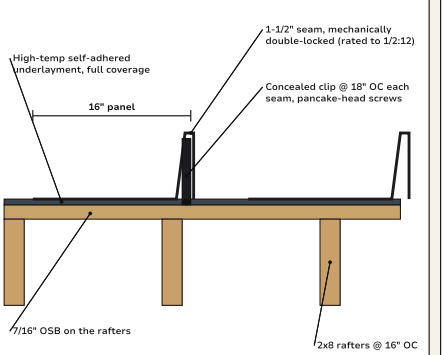

## 3 COVER POST BASE + DECK PENETRATION



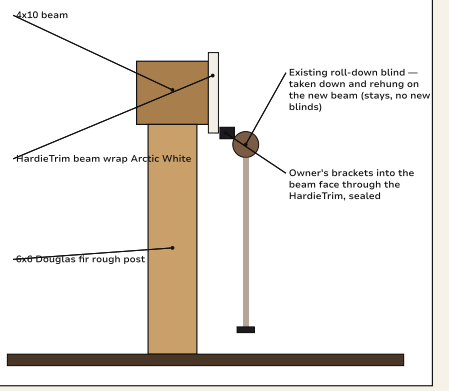

## 4 RAKE EDGE — SECTION ACROSS THE SLOPE



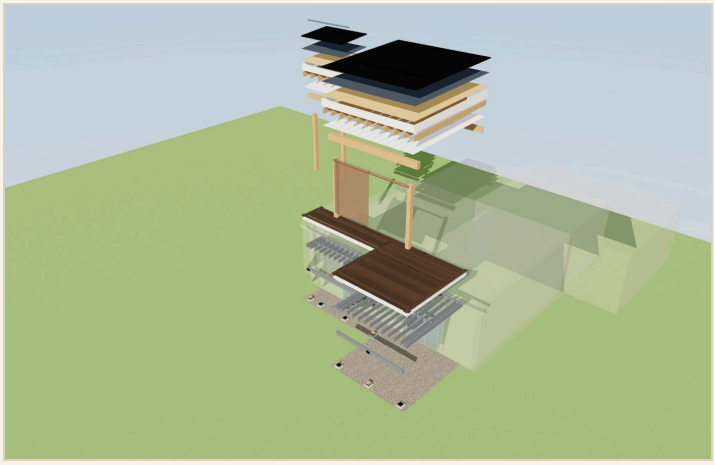

## 5 STANDING SEAM PANEL + CLIP



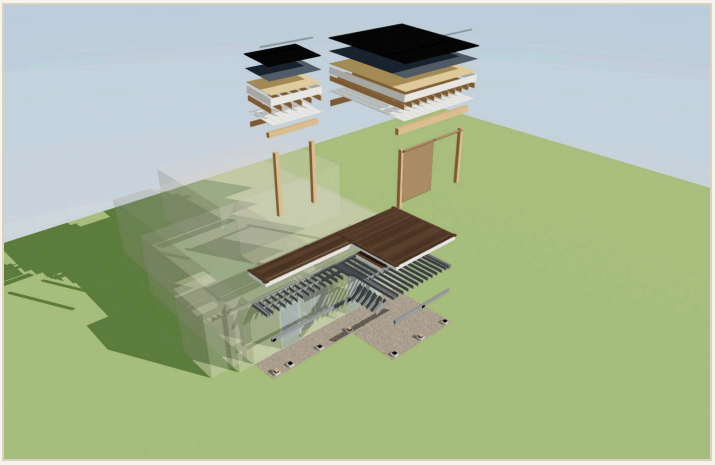

## 6 EXISTING BLINDS AT THE BEAM



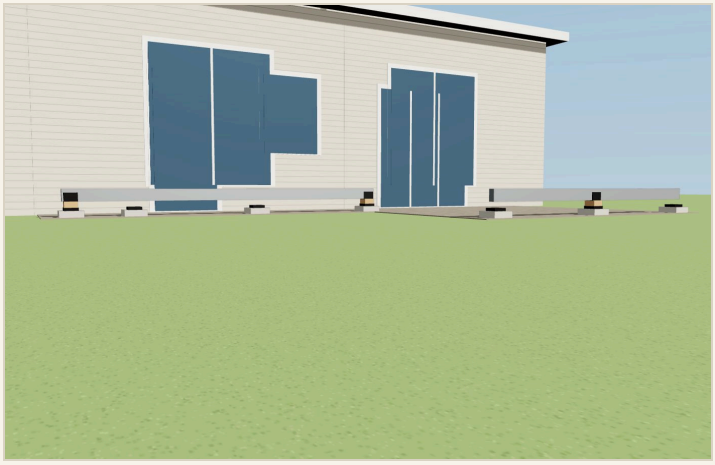
## EXPLODED VIEWS



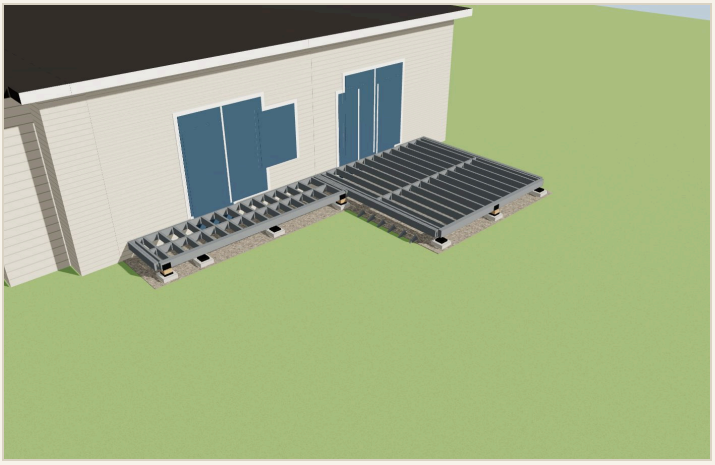
Exploded — every layer



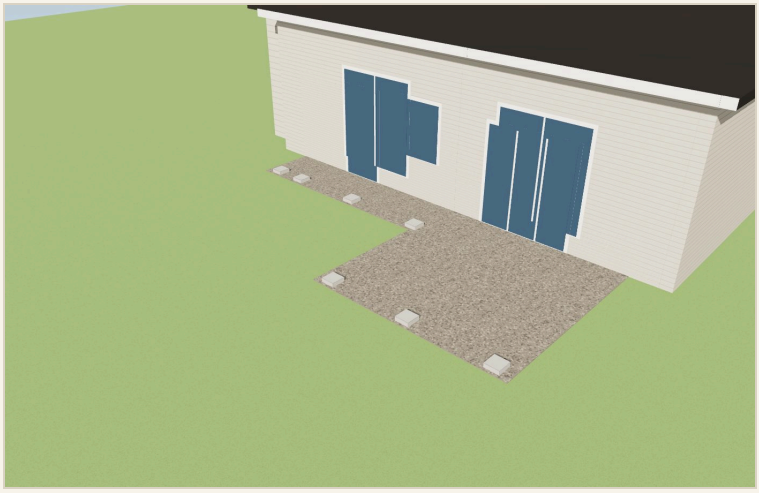
Exploded — from the open corner



Posts and beams on the footings



Frame only — joists, rims, beams, ledger



Step 1 · Footings — 7 Diamond Pier DP-50/50 driven-pin footings

## T-400 · Step 1 · Footings

### GOES IN

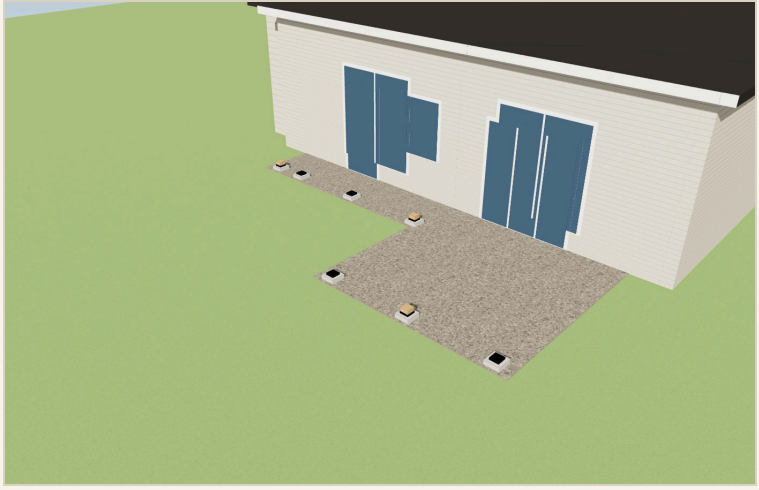
- (7) Diamond Pier DP-50/50 (head + 4 x 50" pins, plugs, caps)

### HOW

- Call 811 and locate before any hole or pin. Strip the existing deck first.
- Pull the beam lines from the house wall: FLUSH boxed 600S162-54 W at 3'-6 3/8" from the reference wall, posts at 7 3/16" / 3'-0" / 8'-4" / 14'-1 3/16"; FLUSH boxed 600S162-54 M at 11'-10 3/8" from the reference wall, posts at 15'-2" / 20'-8" / 26'-2".
- Set 7 Diamond Pier heads level at grade on undisturbed soil and drive the 4 pins in each; caps on.

### CHECK

- Footings on the layout within 1/2"; bearing on undisturbed soil
- Pins fully driven, heads level
- Photos before backfill



Step 2 · Posts — 6x6 Douglas fir rough deck posts on ABA66RZ bases, steel saddles to the beams; 6x6 Douglas fir rough cover posts carry the deck beams where they line up

## T-401 · Step 2 · Posts

### GOES IN

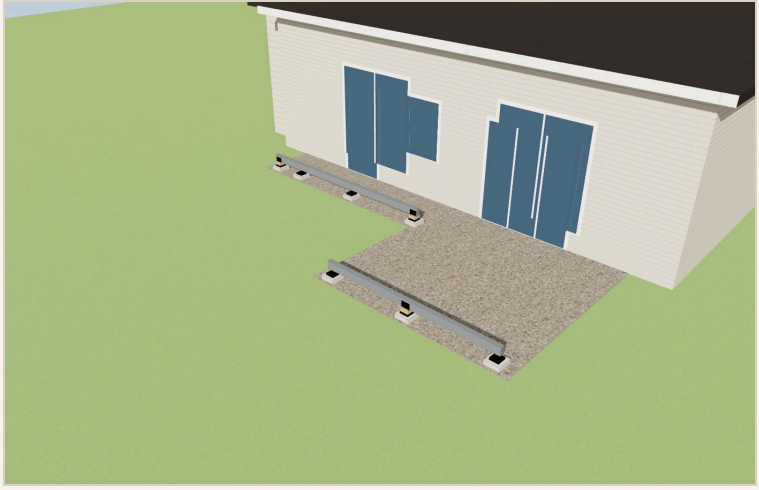
- (4) Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier)
- (1) 6x6x8 Douglas fir rough — deck posts on the deck-only piers
- (3) Simpson ABA66RZ post base on the Diamond Pier bolt — deck post
- (3) Galvanized steel beam saddle, 11 ga — boxed steel beam to the 6x6 post top
- (2) Simpson ABA66RZ post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Main cover
- (2) Simpson ABA66RZ post base on the shared Diamond Pier bolt — cover post (pier counted with the deck) — Walkway cover

### HOW

- Deck-only piers: 6x6 Douglas fir rough posts on ABA66RZ bases, galvanized steel saddles on top carrying the boxed beams, cut so the beam top lands at the joist top (flush beams): FLUSH boxed 600S162-54 W 6" / FLUSH boxed 600S162-54 M 6" — field-measure each.
- Where a roof-cover post lands on the deck beam line it stands on the same Diamond Pier on an ABA66RZ base; the 6x6 Douglas fir rough post runs up to the cover beam and the deck beam bolts to it with the steel side bracket.
- Plumb both ways and brace until the beams are on.

### CHECK

- Posts plumb to a string; tops on the beam line
- Every bracket and saddle fastened: thru-bolts and #12 screws per the engineer



Step 4 · Beams — FLUSH boxed 600S162-54 W (4 posts) · FLUSH boxed 600S162-54 M (3 posts)

## T-403 · Step 4 · Beams

### GOES IN

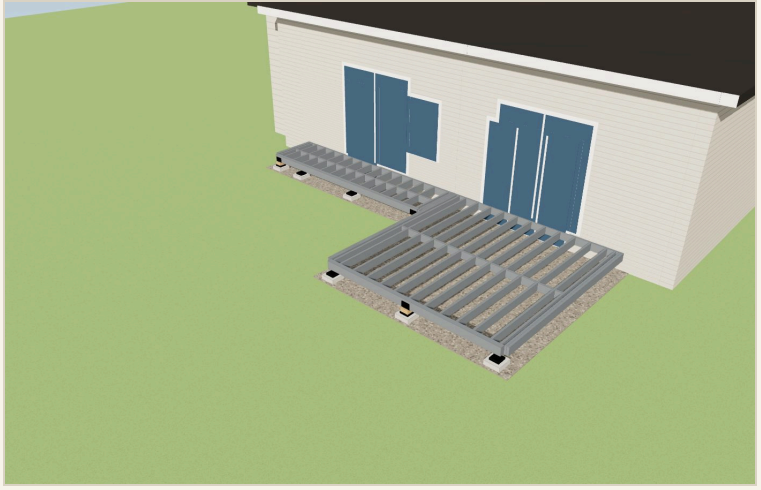
- (51) CEMCO 600S162-54 C, G90 — boxed beams (2 C toe-to-toe)
- (4) Steel side bracket, 11 ga galvanized — deck beam bolted to the 6x6 cover post (shared pier)

### HOW

- FLUSH boxed 600S162-54 W: 14'-1 1/2" one piece, 4 posts @ 5'-9 3/16" max
- FLUSH boxed 600S162-54 M: 11'-5 1/2" one piece, 3 posts @ 5'-6" max
- Box two C sections toe-to-toe, #12 self-drilling screws per the engineer's pattern; set flush with the joist tops; brackets to the stubs and cover posts.

### CHECK

- Beam top = joist top (flush)
- Brackets fully fastened; cut ends cold-galvanized



Step 5 · Frame — CEMCO 600S162-54 C-joists at 12" OC, 600T track rims, ledger track on a steel angle, clip angles, all G90

## T-404 · Step 5 · Frame

### GOES IN

- (347) CEMCO 600S162-54 C-joist (6", 54 mil / 16 ga, G90) — joists, PF joists, blocking
- (77) CEMCO 600T track (54 mil / 16 ga, G90) — rims (single ply, caps the joist ends) + ledger track
- (27) Steel ledger angle 3" x 3" 14 ga G90 — joist seats on the ledger track
- (1) Grace Vycor Deck Protector 12" x 75' (wall membrane behind ledger + side rims)
- (3) Aluminum L-flashing 10' (over ledger + side rims)
- (1) 6-mil poly sheeting 10' x 100' — under the decks
- (1.3) 3/4" washed gravel, 2" over the poly — under the decks (delivered)
- (56) Steel clip angle 1-1/2" x 1-1/2" x 7" 16 ga G90 — joist to track / beam
- (3) #12-14 x 1" hex-washer self-drilling screw, Climaseal, 500 ct
- (1) Structural screw 1/4" x 3-1/2" through the ledger track into the house rim board (or per the engineer)
- (8) Lateral-load tension tie for steel joists (Simpson DTT2Z w/ steel-joist kit or the engineer's detail) — IRC R507.9.2 equivalent

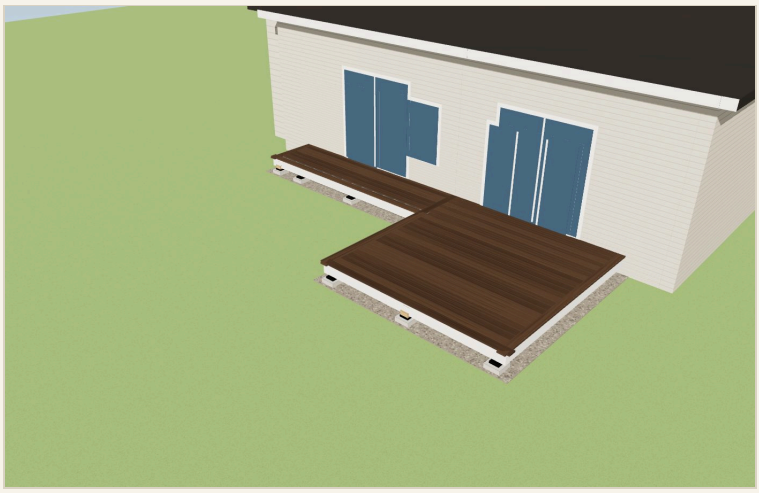
### HOW

- 600T ledger track + 3x3 ledger angle, 1/4" structural screws 2 rows staggered 16" OC into the rim board (engineer confirms) · flashing and membrane as for wood
- W: 15 CEMCO 600S162-54 C-joists @ 12" OC x 3'-3 1/8"; M: 13 CEMCO 600S162-54 C-joists @ 12" OC x 11'-7 1/8". Joists seat on the ledger angle and clip to the ledger track and the boxed beams with clip angles, 4 #12 screws per leg.
- Track caps every joist end at the rims; picture-frame and breaker joists doubled where the borders land.
- Gravel over 6-mil poly under the frame before the decking (2" min. clear at the walkway low end).
- 600S162-54 C-joists @ 12" OC · single-ply 600T track rims · boxed 600S162-54 beams · all G90 · clip angles with #12 self-drilling screws (4 per leg) · cut ends cold-galvanized · no joist tape needed

### CHECK

- Frame square: diagonals within 1/8"
- Every clip fully screwed; ledger screws per the schedule

- Cut ends cold-galvanized; photos of the ledger, clips and footings before decking



Step 6 · Decking — TimberTech Landmark American Walnut, American Walnut border, hidden fasteners

### T-405 · Step 6 · Decking

GOES IN

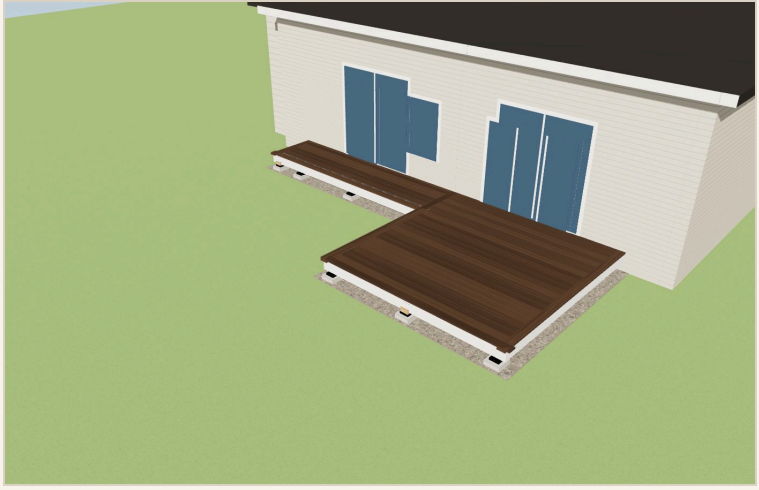
- (25) TimberTech Landmark American Walnut 1x6x12 Grooved
- (7) TimberTech Landmark American Walnut 1x6x16 Grooved
- (4) TimberTech Landmark American Walnut 1x6x16 Square Edge — borders & dividers
- (1) TimberTech Landmark American Walnut 1x6x20 Square Edge — borders & dividers
- (5) Deck skirt — HardieTrim 4/4 x 12 x 12' smooth ColorPlus Arctic White (wraps the double rim, in place of the composite fascia)
- (1) Color-matched trim screws for HardieTrim, 100 ct (Arctic White)
- (1) ColorPlus touch-up kit, Arctic White

HOW

- CONCEALoc at every joist in every gap (437)
- Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then alternate top/bottom every 12", pre-drilled · stair treads 2 per stringer per board
- Rows per panel: panel 1 (0"–14'-8") 7 rows · panel 2 (14'-8"–26'-8") 25 rows · 2 ripped pieces · 1 breaker. Borders and breakers first, then the field rows from the outer edge toward the house.

CHECK

- 3/16" gaps held
- No cut end shows at the outer edge
- Color: every board from the same run



Step 7 · (no rail)

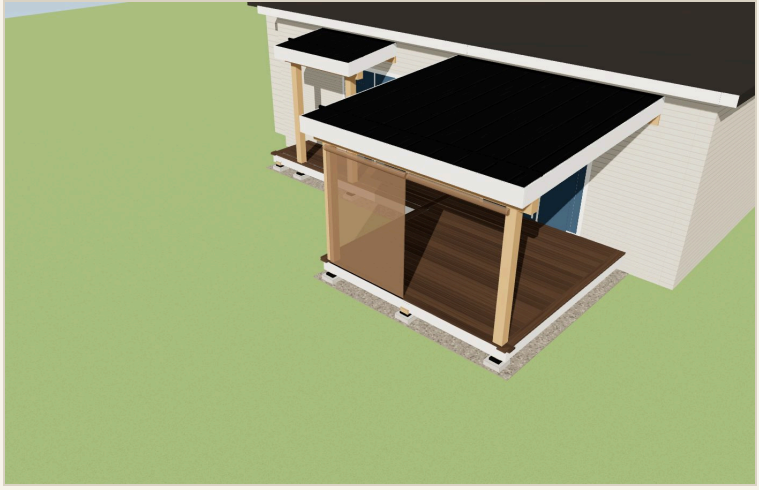
## T-406 · Step 7 · Rail

GOES IN

HOW

CHECK

- —
- Cables tensioned / balusters under 4" clear
- End of construction



Step 8 · Roof covers — 6x6 Douglas fir posts, 4x10 beam, 2x8 rafters, OSB, underlayment, standing seam, Hardie soffit / fascia / rakes

## T-407 · Step 8 · Porch cover

### GOES IN

- (2) 6x6x10 Douglas fir rough post (cover) — Main cover
- (1) 4x10x14 Douglas fir — cover beam — Main cover
- (1) 2x8x16 #1 SYP — cover ledger on the house — Main cover
- (10) 2x8x14 #1 SYP — cover rafters @ 16" OC — Main cover
- (3) 2x8x16 #1 SYP — cover fascia / sub-fascia — Main cover
- (6) 7/16" OSB roof sheathing 4x8 — Main cover
- (170) Standing seam roof — 24 ga steel 16" 1-1/2" mechanically seamed (double-lock, low-slope rated) panels x 13.2' (Black, PVDF), 10 panels — Main cover
- (1) High-temp self-adhered ice & water underlayment, 2 sq roll (under every metal panel) — Main cover
- (2) Standing seam eave / drip trim 10', color-matched — Main cover
- (3) Standing seam gable / rake trim 10', color-matched — Main cover
- (2) Headwall (ledger) flashing with Z-closure 10', color-matched — into the house siding / counterflash — Main cover
- (3) Foam + metal closure strips at eave and headwall, 10' — Main cover
- (1) Panel clips + pancake-head screws, box of 250 — Main cover
- (1) Color-matched #10 woodgrip trim screws w/ washers, 250 — Main cover

### HOW

- Cover ledger on the house at the top of the pitch, LedgerLOK 2 rows; 6x6 Douglas fir rough posts on ABA66RZ bases on the Diamond Piers (shared with the deck beam where they line up), main cover: 4x10 beam on BC46Z caps / walkway cover: (2)2x8 beam on AC6Z caps; 2x8 rafters @ 16" OC with H2.5AZ ties and the overhang; 2x8 sub-fascia and OSB. Cover framing is wood (SYP 2x8 rafters, Douglas fir posts; Douglas fir 4x beam).
- High-temp self-adhered underlayment over every sheet, headwall flashing counterflashed into the house wall, eave trim, then the 24 ga snap-lock panels one piece eave to headwall — clips 18" OC on each seam, no face screws in the field; rake trim and closures last; snow retention bar clamped to

the seams 12-18" above the low eave, the full width.

- Ceiling: HardieSoffit 4x10 cedarmill Arctic White panels 6" OC stainless ring-shank at every rafter; HardieTrim 4/4 x 7.25 smooth Arctic White fascia and rakes over the 2x8 sub-fascia, color-matched trim screws.

### CHECK

- Ledger flashed and counterflashed — hose test before the ceiling closes
- Posts plumb over blocking, beam level, rafters tied
- Panels straight, seams engaged full length, no oil-canning; snow bar continuous
- Water sheds clear of the deck edge

## 3D MODEL



The interactive 3D model lives at the online version of this document.



Quantities from the drawings; the stamped engineering set governs structure. Quote requested — no pricing on this sheet.